

S2PX / Scan2Plan CPQ

Handoff Package for Elijah Tuttle

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Status: Owen declined the S2PX deal on March 25, 2026. This handoff covers infrastructure and data transfer so Elijah can operate the CPQ independently.

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1. What You're Getting (and What You're Not)

Transfers to Elijah

- **GCP project access** (s2px-production) — IAM roles for managing infrastructure
- **GCS scan data and project files** — all business data in the storage buckets
- **Database access** — Cloud SQL connection for the production PostgreSQL instance
- **Domain** (scan2plan.io) — can be transferred if desired
- **Product documentation and architecture docs** — this document and related specs
- **Firebase Hosting sites** — the deployed frontend URLs

Does NOT Transfer

- **Source code** — The CPTV27/S2PX GitHub repository is Chase's. It is NOT being transferred. No codebase leaves.
- **GitHub access** — All collaborator access was revoked on March 25, 2026.
- **Any HDX / Measurably Better / Hillbilly Dreams IP** — completely separate platform.
- If you need the CPQ code, that's a separate conversation about licensing.

2. Product Overview — What the CPQ Does

The S2PX CPQ is a full-lifecycle platform for 3D scanning and BIM businesses:

Feature	Description
Sales Pipeline	Kanban board / lead management with stage tracking
Scoping Forms	135-field intake form with geocoding, multi-building support
Deal Workspace	Full lead detail view with quote builder
CPQ Calculator	Multi-building, multi-discipline pricing engine with margin controls
Production Pipeline	Project tracking from scan through delivery
Proposal Builder	PDF proposal generation with template system
Signature Flow	E-signature + post-signature email notification
Scan Tech Portal	Field technician mobile interface with photo analysis AI
QuickBooks Integration	Estimate/invoice sync via OAuth2
Google Drive Integration	File management via GCS 4-bucket architecture

Feature	Description
Knowledge Base	Internal wiki with AI Q&A; chat
Strategy Center	CEO strategy drafts with 3-tier governance
AI Agents (7)	Pricing, Operator, Photo Analysis, Checklist, Audio-to-Scoping, KB Chat, Strategy Advisor
Scorecard	Company performance metrics and financial dashboards
Website Manager	Content management for scan2plan.io marketing site

Note on the CPQ Calculator

The pricing engine was the single most labor-intensive component of the system. The underlying pricing logic changed repeatedly during development, requiring multiple full refactors of the calculator — a process that consumed days of engineering time per iteration. The current engine (shared/engine/pricingV2.ts) is the stabilized version, backed by database-driven rate tables (15 pricing tables in shared/schema/pricing.ts) and covered by 261 unit tests. The test suite exists specifically because of how many regressions the calculator changes introduced.

3. GCP Project Summary

Field	Value
Project ID	s2px-production
Owner Account	chase@scan2plan.io
Service Account	s2px-api@s2px-production.iam.gserviceaccount.com
Region (primary)	us-east4 (N. Virginia)
Region (sandbox)	us-central1 (Iowa)
Console URL	console.cloud.google.com/home/dashboard?project=s2px-production

Services in Use

- Cloud SQL (PostgreSQL)
- Cloud Run (API backend)
- Cloud Storage (4 buckets)
- Firebase Hosting (3 targets)
- Firebase Auth (Google OAuth)
- Secret Manager

- Cloud Tasks
- Cloud CDN (signed URLs)

4. Google Cloud Storage — Buckets & Data

Four-bucket architecture:

Bucket	Purpose	Env Var
s2p-core-vault	Primary storage — proposals, certificates, exports	GCS_BUCKET
s2p-active-projects	Active project data — point clouds, virtual tours, deliverables	GCS_ACTIVE_BUCKET
s2p-incoming-staging	Upload staging area — files land here before processing	GCS_STAGING_BUCKET
s2p-quarantine	Quarantine bucket for failed/suspicious uploads	(security allowlist)

What's in the Buckets

- Scan data (.e57, .las, .laz point cloud files)
- Virtual tour assets (equirectangular panoramas, floor plan SVGs)
- Proposals (generated PDFs)
- Certificates of Assurance (generated PDFs)
- Field photos (scan tech uploads)
- Project deliverables (BIM files, CAD exports)

All of this is Scan2Plan business data and can be transferred.

GCS Folder Structure (inside s2p-active-projects)

Each production project follows this canonical folder layout:

```
{ProjectFolder}/
project.json (auto-generated sidecar metadata)
Deliverables/
Point Cloud/
{raw-files} (.las, .laz, .e57, .rcp, .rcs)
pointree/ (auto-generated by point cloud worker)
metadata.json, octree.bin, hierarchy.bin
Virtual Tour/
tour-manifest.json (auto-generated by virtual tour worker)
floorplan.svg
panoramas/ (001.ipa, 002.ipa, ...)
BIM/CAD/
```

Photos/
Documents/

Scoping form uploads land in a different path inside s2p-core-vault:

scoping/{UPID}/{fieldName}/{filename}

Auto-Processing Pipelines

Files uploaded to GCS don't just sit there — two Cloud Run workers process them automatically:

Worker	Input	Output	Trigger
Virtual Tour	ZIP bundle (DWG + CSV + panoramas)	tour-manifest.json + SVG + panoramas	POST /api/projects/:id/virtual-tour/process
Point Cloud	Raw .e57/.las/.laz files	Potree octree (metadata.json, octree.bin, hierarchy.bin)	Manual worker invocation
Sidecar Writer	Production data changes	project.json metadata per project folder	Automatic on stage transitions, uploads, quotes

File Type Categories (used by sidecar writer)

Category	Extensions
Point Cloud	e57, las, laz, pts, ptx, xyz, ply, pcd, rcp, rcs
BIM/CAD	rvt, rfa, dwg, dxf, ifc, nwd, nwc, skp, 3dm
Photos	jpg, jpeg, png, tif, tiff, bmp, heic, raw, cr2, nef
Archives	zip, rar, 7z, tar, gz
Documents	pdf, doc, docx, xls, xlsx, csv, txt, rtf

5. Firebase Projects & Hosting

Field	Value
Project ID	s2px-production
Console	console.firebase.google.com/project/s2px-production

Hosting Targets

Target	Firebase Site	URL	API Service
production	s2px	s2px.web.app	s2px-api (us-east4)
sandbox	s2px-sandbox	s2px-sandbox.web.app	s2px-api-sandbox (us-central1)
staging	s2px-staging	s2px-staging.web.app	s2px-api-staging (us-east4)

Firestore Services Used

- **Auth** — Google OAuth provider (this is the only auth method)
- **Hosting** — Static SPA hosting with API rewrites to Cloud Run
- **Storage Rules** — defined in storage.rules
- **Firestore** — Client-side only, supplementary to Cloud SQL (see below)

Firestore Collections (Not Finalized)

Firestore was being evaluated as a supplementary data layer alongside Cloud SQL. This was **not finalized** — the collections below exist in code but their production state should be verified. All Firestore access is client-side only (no server-side reads/writes).

Collection	Purpose	Status
wiki_pages	Knowledge Base articles (legacy)	Active — has security rules
deal_sessions	Viewer session telemetry for Deal Presentations	Not finalized
deal_events	Section enter/exit tracking with dwell time	Not finalized
deal_voice	Voice memo transcriptions for deal proposals	Not finalized
deal_cta_clicks	CTA click tracking	Not finalized

Note: Only wiki_pages has explicit Firestore security rules (requires auth). The deal_* collections have no explicit rules — they fall under the default deny rule. The primary database for all business data is Cloud SQL (PostgreSQL).

■ **WARNING: All three environments share the same Firebase Auth project. There is no auth isolation. Destructive auth operations in one environment affect all others.**

6. Database

Field	Value
Type	PostgreSQL (Google Cloud SQL)
Instance	s2px-production-db
ORM	Drizzle ORM
Connection	Via DATABASE_URL env var
Max Connections	~100 (Cloud SQL instance setting)
Pool Size	5 per Cloud Run instance
Backups	Daily automated, 7-day retention + PITR enabled

Schema Files (~90 tables total)

- shared/schema/db.ts — 59 core tables (users, scoping_forms, quotes, proposals, production, etc.)
- shared/schema/pricing.ts — 15 pricing engine tables (rate tables, baselines, modifiers)
- shared/schema/platform.ts — 2 tables (content, roadmap)
- shared/schema/market-intelligence.ts — 9 tables (segments, competitors, scanners, benchmarks)
- shared/schema/initiatives.ts — 1 table
- Plus: deal-arena.ts, engagement.ts, aiUsage.ts

Key Tables

Table	Purpose
users	Firestore-authed users with RBAC roles
scoping_forms	Deal intake (135 columns)
quotes	Generated quotes with line items
proposals	Proposal documents with PDF generation metadata
production_projects	Active production pipeline
qbo_tokens	QuickBooks OAuth tokens
kb_sections	Knowledge base content
strategy_drafts	CEO strategy documents
pricing_constants	V2 pricing engine config
scan_checklists	Field technician QC checklists

Migrations

- 18 migration files in migrations/ directory
- Managed via Drizzle Kit (npm run db:generate, npm run db:migrate)

7. Cloud Run Services (API Backend)

Service	Region	Purpose
s2px-api	us-east4	Production API
s2px-api-sandbox	us-central1	Sandbox API
s2px-api-staging	us-east4	Staging API

Production Configuration

Setting	Value
Max instances	10 (capped for connection safety)
Concurrency	80
Pool max	5 per instance

Connection pool rule: $\text{instances} \times \text{pool_max} \leq 0.7 \times \text{cloud_sql_max_connections}$

8. Environment Variables & Secrets

Every env var the S2PX app needs. Values are **not included** — Chase will provide credentials separately.

Required

Variable	Purpose
DATABASE_URL	PostgreSQL connection string (Cloud SQL)
DB_POOL_MAX	Connection pool max (currently 5)
DB_POOL_MIN	Connection pool min (currently 1)
GCS_BUCKET	Primary GCS bucket (s2p-core-vault)
VITE_FIREBASE_API_KEY	Firebase Web API key

Variable	Purpose
VITE_FIREBASE_AUTH_DOMAIN	Firebase auth domain
VITE_FIREBASE_PROJECT_ID	Firebase project ID
VITE_FIREBASE_STORAGE_BUCKET	Firebase storage bucket
VITE_FIREBASE_MESSAGING_SENDER_ID	Firebase messaging sender ID
VITE_FIREBASE_APP_ID	Firebase app ID
VITE_FIREBASE_MEASUREMENT_ID	Firebase analytics measurement ID
FIREBASE_PROJECT_ID	Server-side Firebase project ID
GEMINI_API_KEY	Google Gemini AI (powers all 7 AI agents)
GOOGLE_MAPS_API_KEY	Server-side Maps API (geocoding, distance)
VITE_GOOGLE_MAPS_API_KEY	Client-side Maps API

Optional

Variable	Purpose
ANTHROPIC_API_KEY	Claude AI (KB extraction, notebook)
XAI_API_KEY	Grok (sentiment analysis, strategy)
QBO_CLIENT_ID	QuickBooks Online OAuth2 client ID
QBO_CLIENT_SECRET	QuickBooks Online OAuth2 client secret
QBO_REDIRECT_URI	QuickBooks OAuth callback URL
QBO_ENVIRONMENT	sandbox or production
SENDGRID_API_KEY	Email delivery (proposals)
SENDGRID_FROM_EMAIL	Sender address for proposals
ENCRYPTION_KEY	Encrypts stored API keys in DB
GCS_ACTIVE_BUCKET	Active projects bucket (s2p-active-projects)
GCS_STAGING_BUCKET	Upload staging bucket (s2p-incoming-staging)

Where Secrets Are Stored

- **GCP Secret Manager** on s2px-production project

- CDN signing key stored as s2px-cdn-key in Secret Manager
- Cloud Run services pull secrets from Secret Manager at deploy time

9. Domains & DNS

Domain	Purpose	Registrar	Notes
scan2plan.io	Primary domain	TBD — verify with Chase	Can be transferred
s2px.web.app	Production Firebase Hosting	Firebase-managed	Tied to Firebase project
s2px-sandbox.web.app	Sandbox Firebase Hosting	Firebase-managed	Tied to Firebase project
s2px-staging.web.app	Staging Firebase Hosting	Firebase-managed	Tied to Firebase project

10. Third-Party Integrations

Service	Purpose	Account Owner
Firebase	Auth (Google OAuth) + hosting	chase@scan2plan.io
Google Cloud Storage	4-bucket file architecture	s2px-production project
Google Maps	Address geocoding, travel calc	API key on s2px-production
Google Gemini AI	7 AI agents (server-side only)	API key
QuickBooks Online	Estimate/invoice sync	Owen's QBO account
SendGrid	Proposal email delivery	API key
Google Cloud Tasks	Async job processing	s2px-production project
Google Cloud CDN	Secure file delivery	s2px-production project

■ **QuickBooks Note:** The QBO integration connects to Owen's QuickBooks account. Since Owen is out, this connection will need to be either disconnected (revoke OAuth tokens in qbo_tokens table) or reconnected to a different QBO account.

11. Auth Architecture

- **Provider:** Firebase Auth with Google OAuth (only auth method)

- **Flow:** Client-side Google sign-in → Firebase token → Bearer token in every API call → server/middleware/auth.ts verifies token and attaches req.user
- **Roles:** developer, admin, ceo, production, scantech, viewer
- **Frontend guard:** <RequireAuth allowedRoles={[...]}>
- **Backend guard:** requireRole('role1', 'role2') middleware
- **No password auth** — Google OAuth only

12. How to Get This Running

Prerequisites

- Node.js 20+
- Access to s2px-production GCP project (Chase grants IAM roles)
- gcloud CLI authenticated as an account with project access
- Cloud SQL Auth Proxy (for local development) or a DATABASE_URL connection string

Step 1: Get GCP Access

Chase adds your Google account to the s2px-production project with these IAM roles:

- roles/cloudsql.client (database access)
- roles/storage.admin (GCS bucket access)
- roles/run.admin (Cloud Run management)
- roles/firebase.admin (Firebase hosting + auth)
- roles/secretmanager.secretAccessor (read secrets)

Step 2: Get Credentials

Chase provides:

- DATABASE_URL (Cloud SQL connection string)
- Firebase config values (6 VITE_FIREBASE_* vars)
- GEMINI_API_KEY
- GOOGLE_MAPS_API_KEY (both server and client keys)
- Any optional keys you need (SendGrid, QBO, etc.)

Step 3: Verify Access

```
gcloud auth login
gcloud config set project s2px-production
gcloud sql instances list
gcloud storage buckets list
gcloud run services list
```

```
firebase projects:list
```

Step 4: Database Access

Option A: Cloud SQL Auth Proxy (for local dev)

```
cloud-sql-proxy s2px-production:us-east4:s2px-production-db
```

Option B: Direct connection via DATABASE_URL (Chase provides the connection string)

Step 5: Deploy Updates (if you have the code)

```
firebase deploy --only hosting:production
```

```
gcloud run deploy s2px-api --source . --region us-east4 --project s2px-production
```

Note: Steps 4–5 require the source code, which is not part of this handoff. These are here for reference if a licensing arrangement is made separately.

13. Credentials You'll Need From Chase

Chase fills in and delivers separately (never in this document):

- GCP IAM access granted to Elijah's Google account
- DATABASE_URL connection string
- Firebase config block (6 VITE_FIREBASE_* values)
- FIREBASE_PROJECT_ID
- GEMINI_API_KEY
- GOOGLE_MAPS_API_KEY (server)
- VITE_GOOGLE_MAPS_API_KEY (client)
- ENCRYPTION_KEY (if applicable)
- SENDGRID_API_KEY + SENDGRID_FROM_EMAIL (if email delivery needed)
- ANTHROPIC_API_KEY (if Claude AI features needed)
- Domain registrar transfer (if scan2plan.io is being transferred)
- QBO decision: disconnect Owen's account or reconnect to new account

14. Ownership Boundary

Asset	Owner	Transfers?
Source code (CPTV27/S2PX repo)	Chase Pierson	NO
GitHub repository access	Chase Pierson	NO

Asset	Owner	Transfers?
HDX / Measurably Better / Hillbilly Dreams IP	Chase Pierson / HDX	NO
GCS scan data & project files	Business data	YES — Scan2Plan business data
GCP project access (s2px-production)	Chase (owner)	YES — IAM roles can be granted
Cloud SQL database & data	Business data	YES — connection access granted
Firebase Hosting sites	Tied to GCP project	YES — follows GCP project access
Firebase Auth user accounts	Tied to Firebase project	YES — follows Firebase project access
Domain (scan2plan.io)	Chase	YES — can be transferred if desired
Product documentation	General knowledge	YES — this document and arch docs
API keys (Gemini, Maps, SendGrid)	Account-specific	CONDITIONAL — re-provision under Elijah's accounts

Clear Boundaries

- 1. The code is Chase's.** The CPTV27/S2PX repository is private and stays with Chase. No source code is part of this handoff.
- 2. The data is the business's.** GCS files, database records, and project deliverables are Scan2Plan operational data.
- 3. Infrastructure access is grantable.** Chase can add Elijah to the GCP project with IAM roles without transferring ownership.
- 4. API keys should be re-provisioned.** Elijah should create his own Google Cloud project and API keys long-term, rather than depending on Chase's accounts.

This document was generated from the S2PX codebase configuration, GCP project settings, and architecture documentation. GCS bucket inventory should be verified live with `gcloud storage buckets list` once `chase@scan2plan.io` auth is refreshed.